



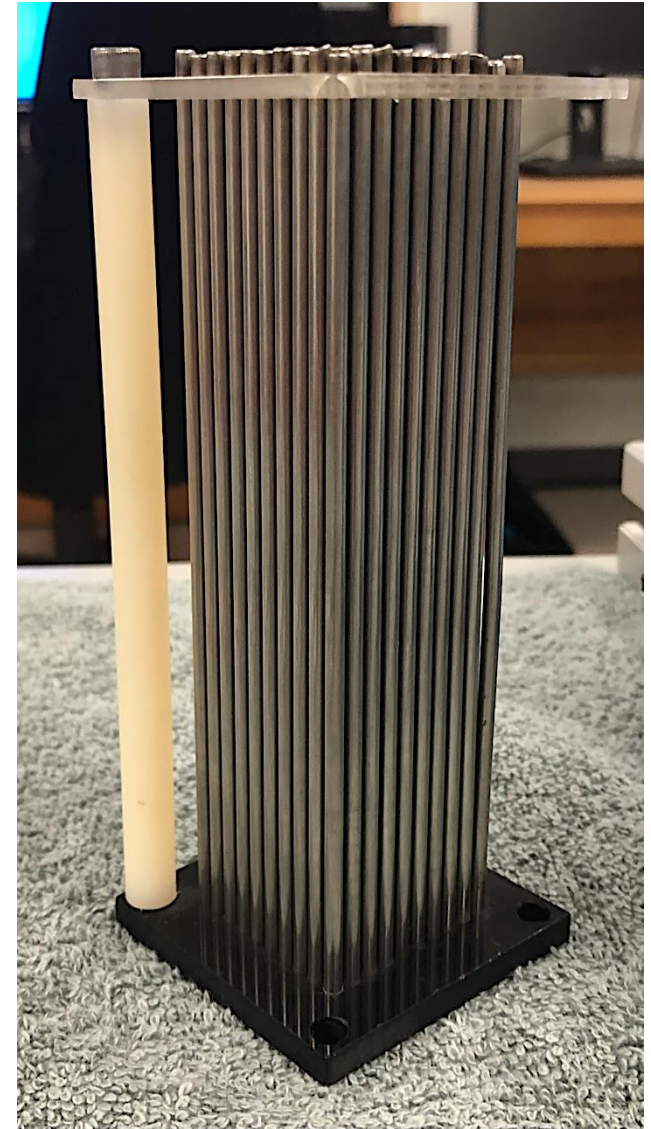
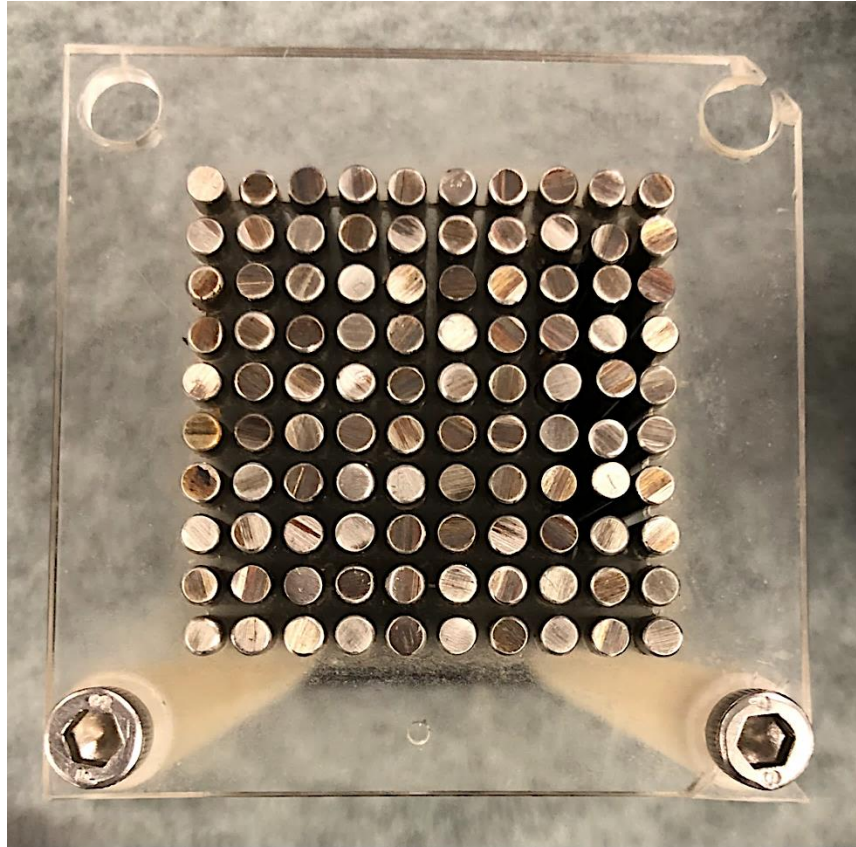
Anderson Localization in a Ferrofluidic Waveguide System with a Topological Defect

Vedant Sundar

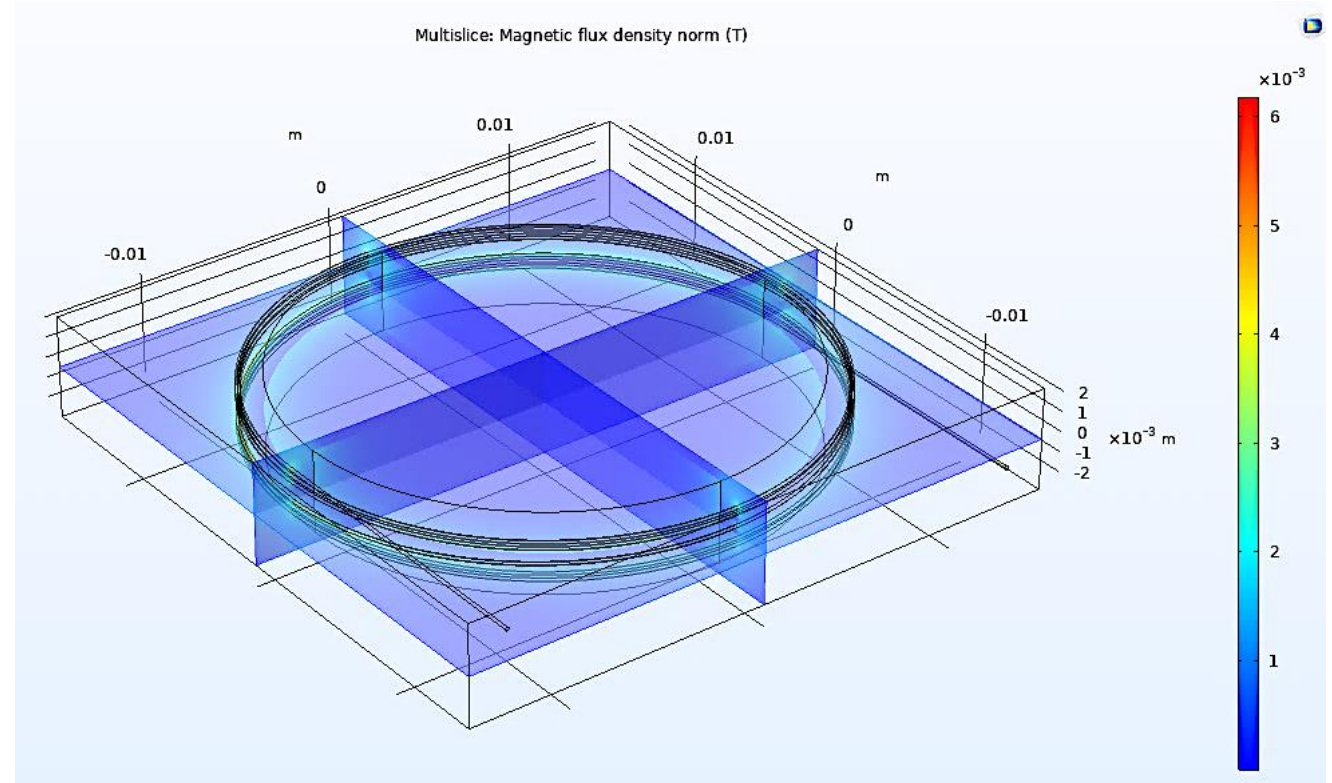
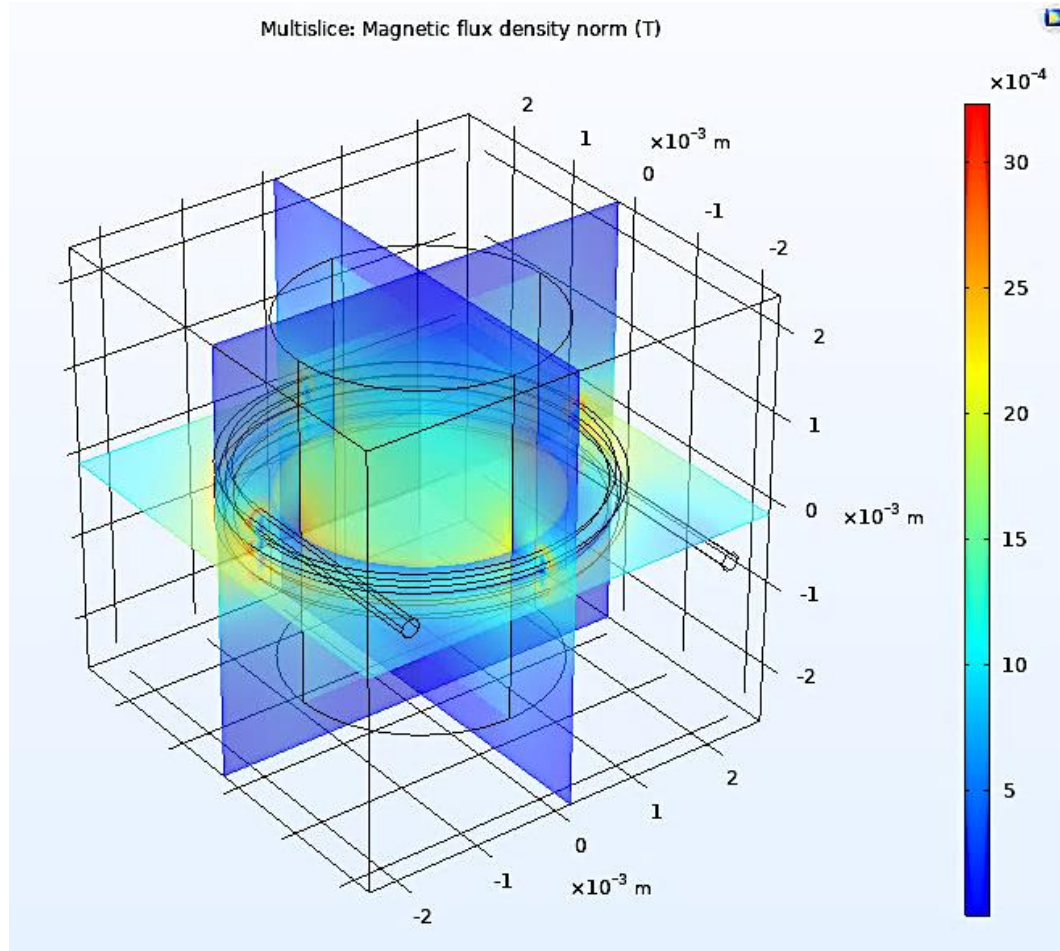
Plano East Sr. High

vedantsundar1@gmail.com

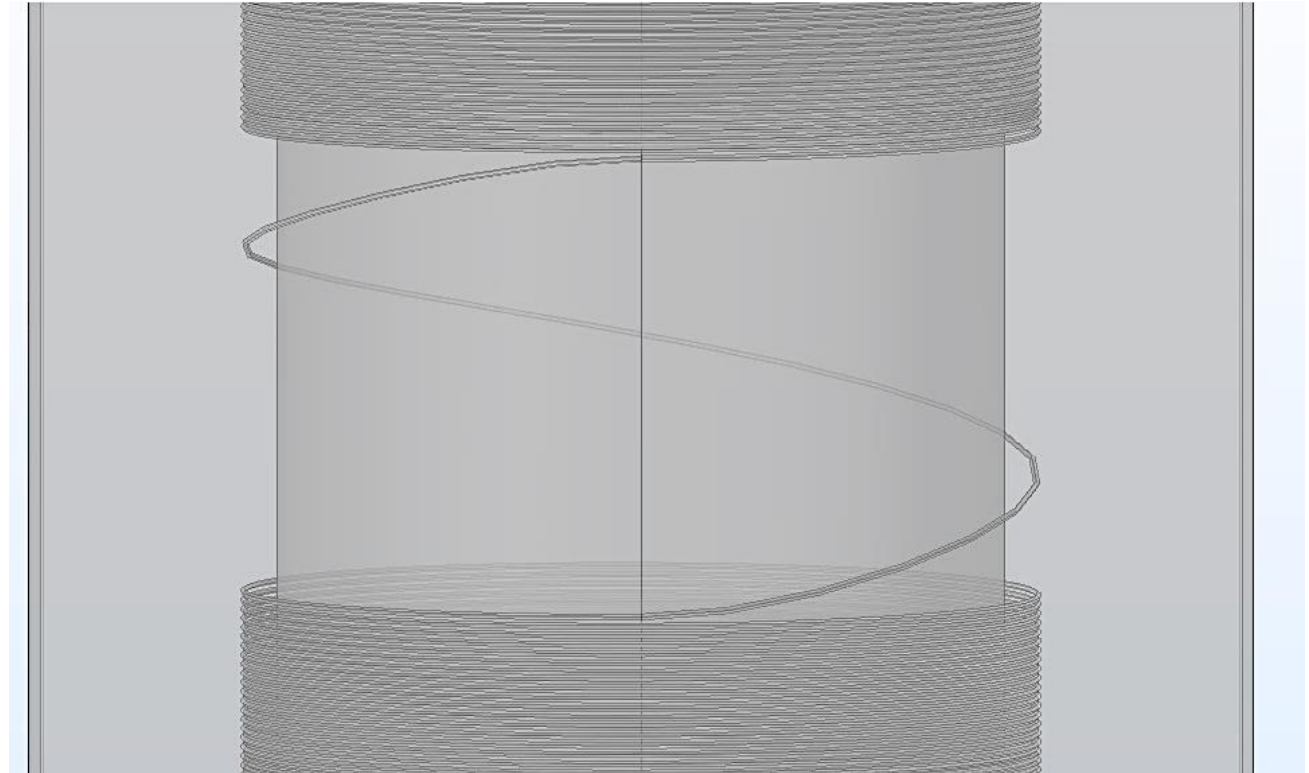
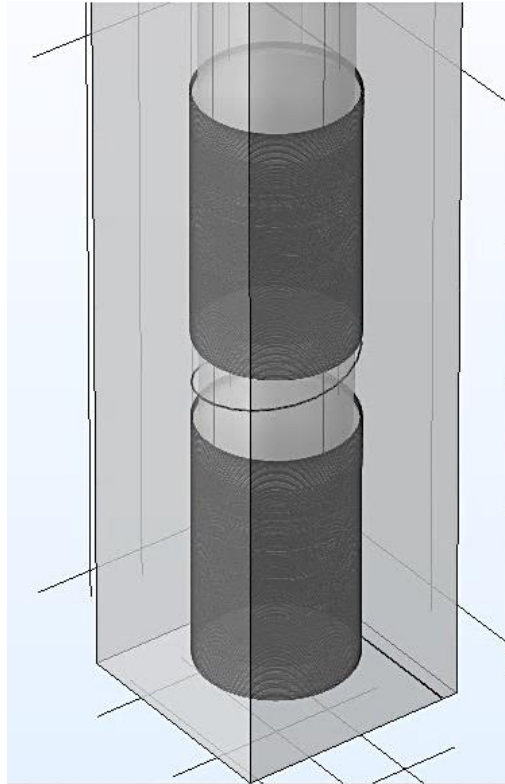
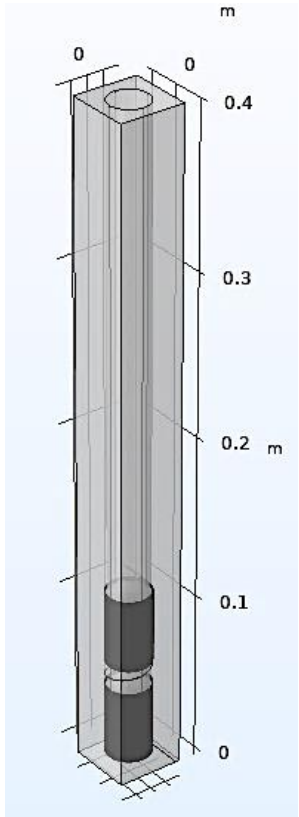
Introduction



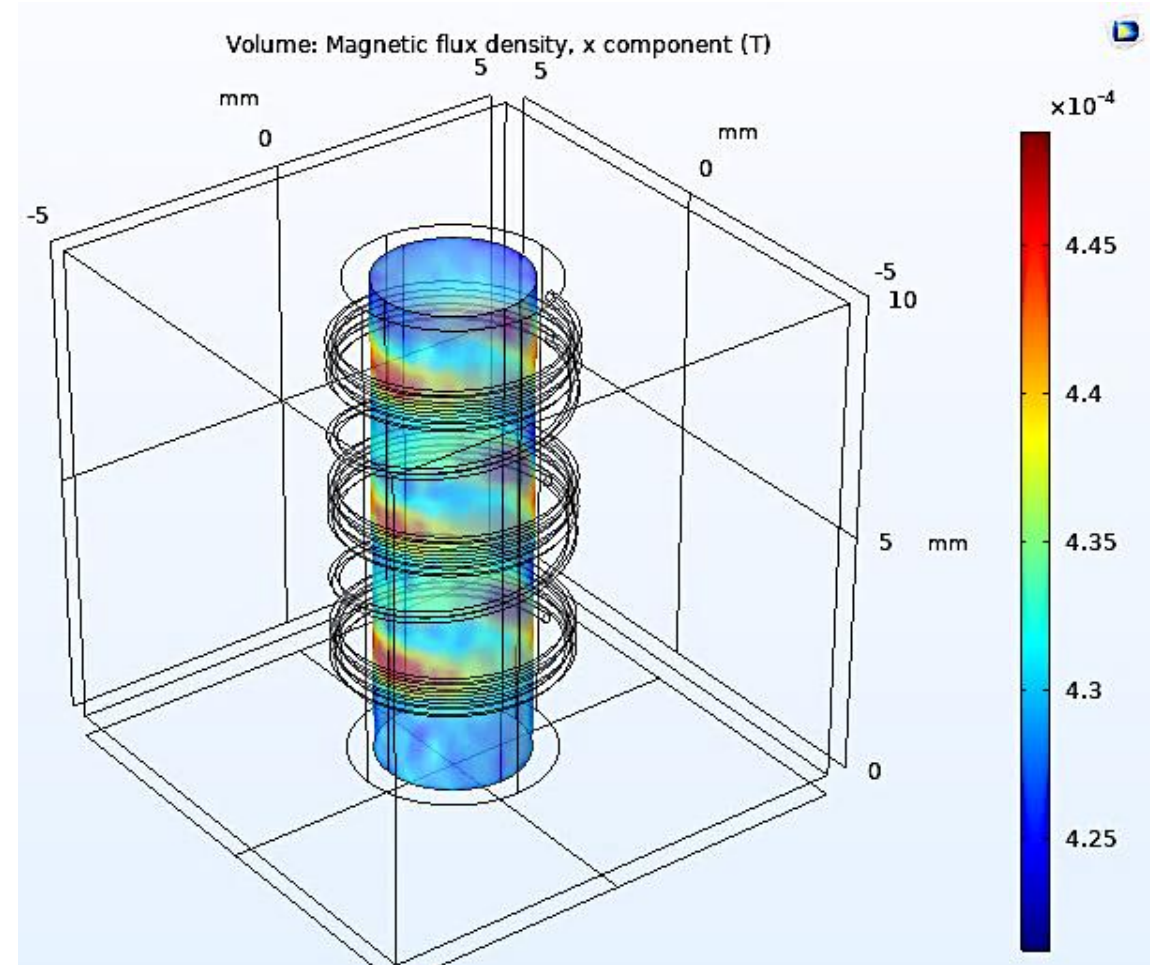
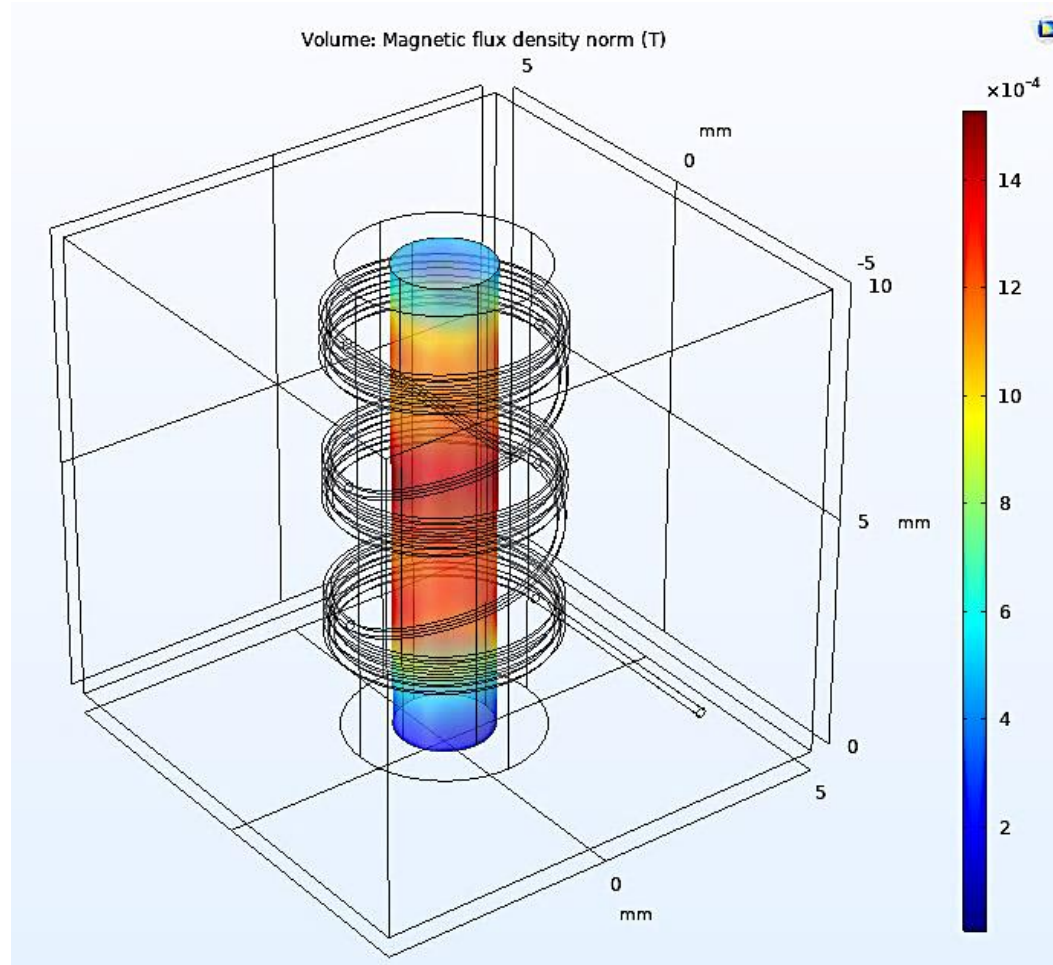
Solenoid Analysis



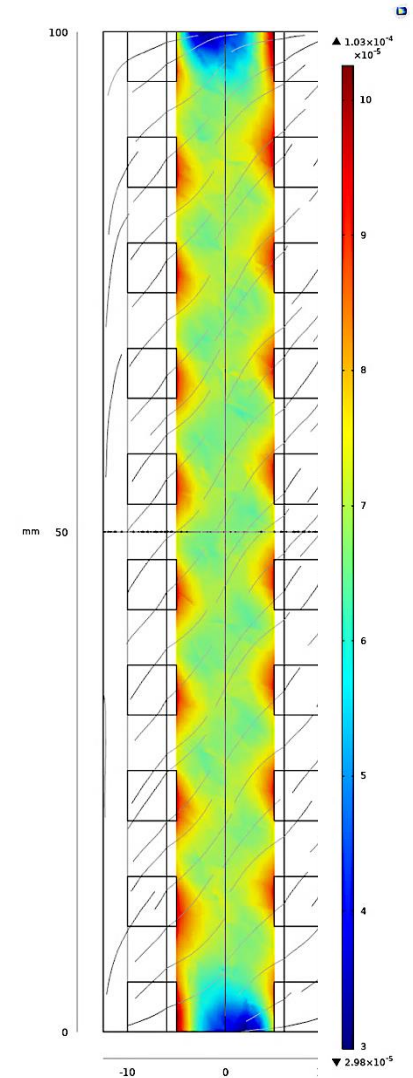
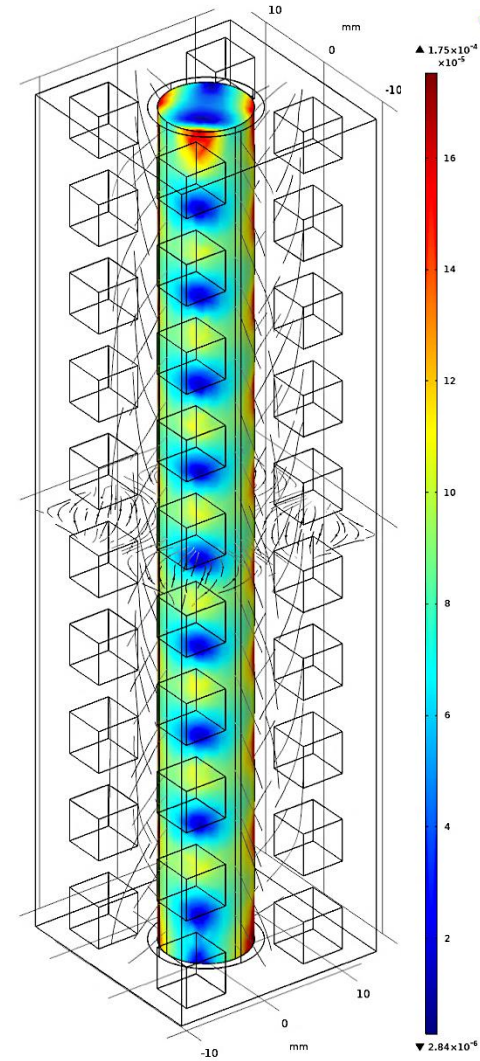
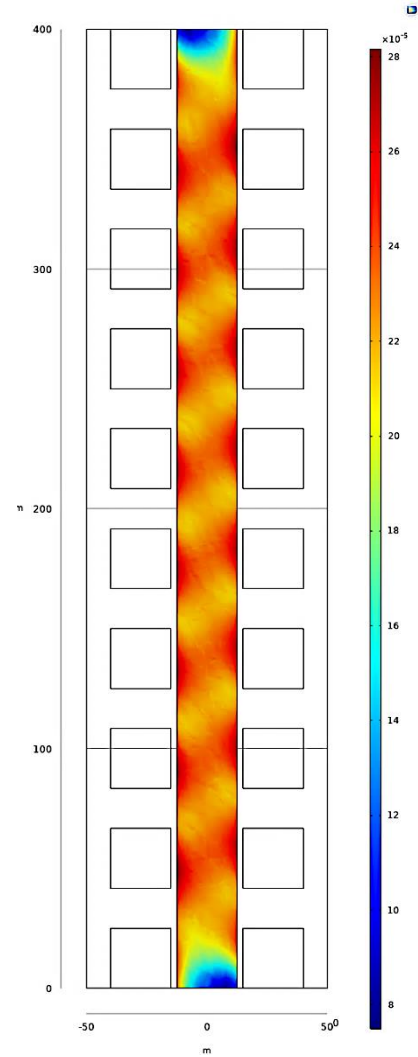
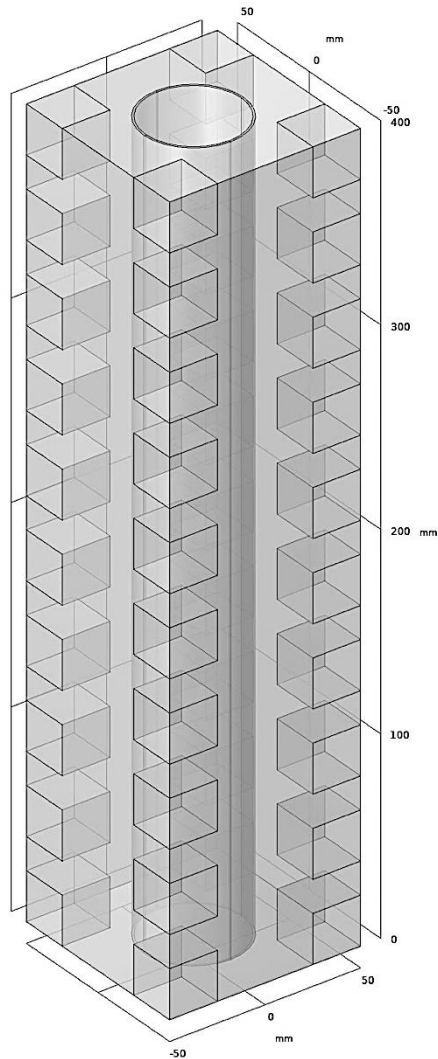
Waveguide Model



Solenoids and Magnets



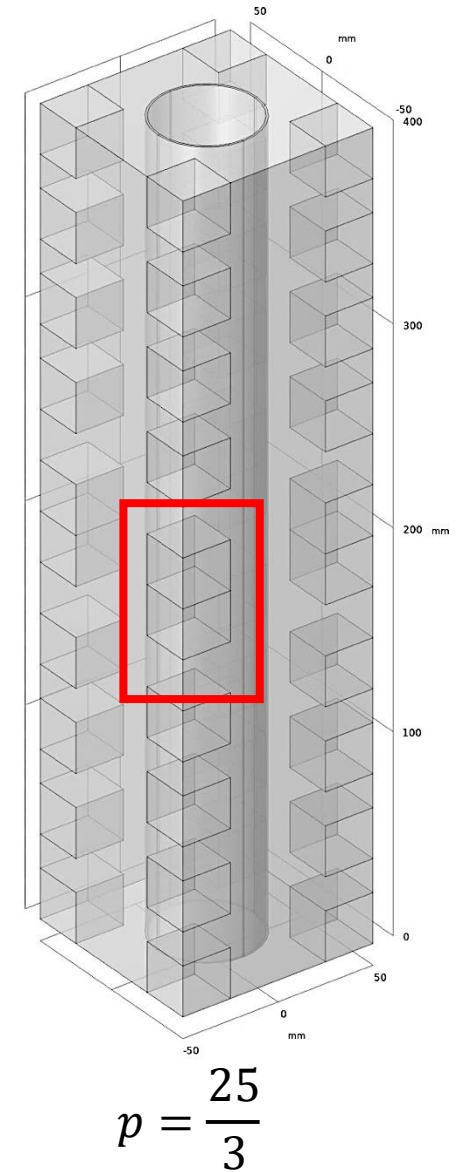
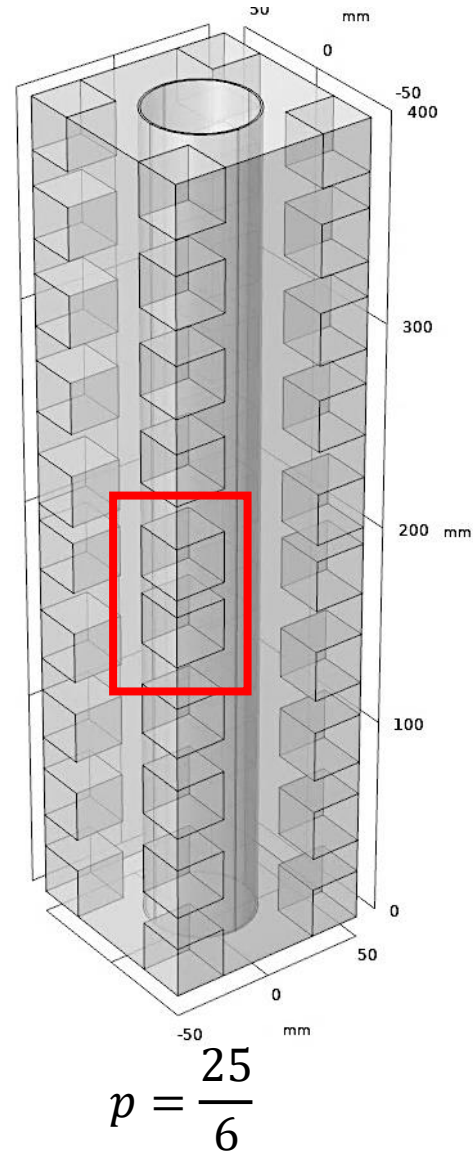
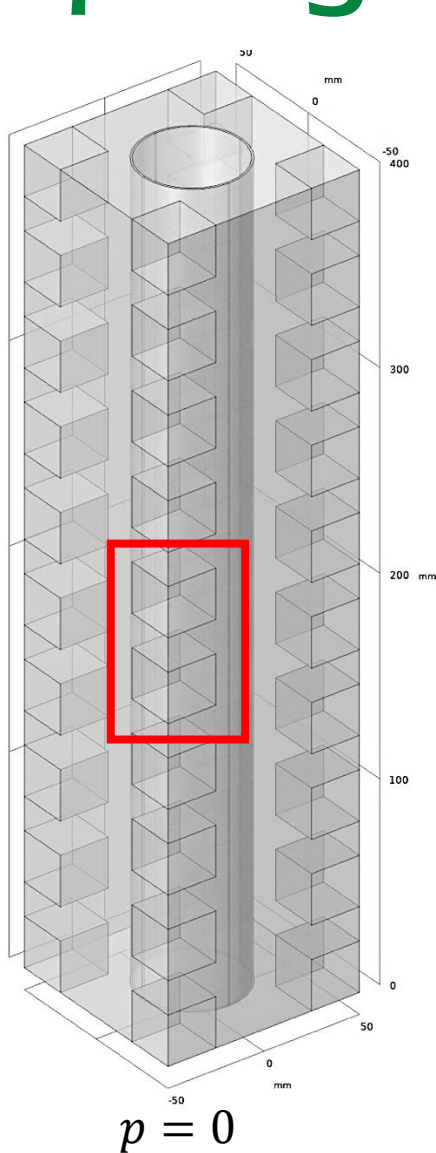
Using N52 Magnets



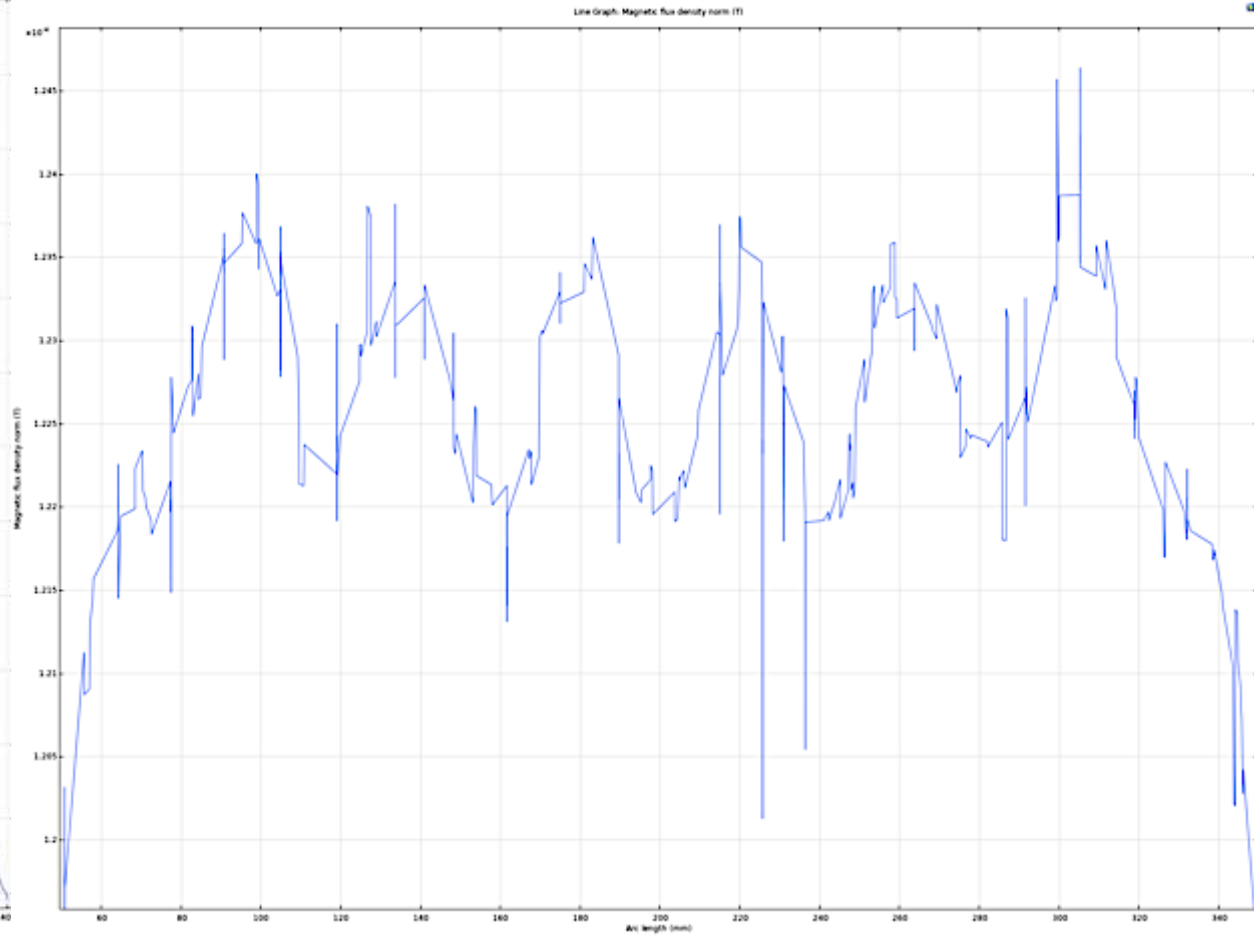
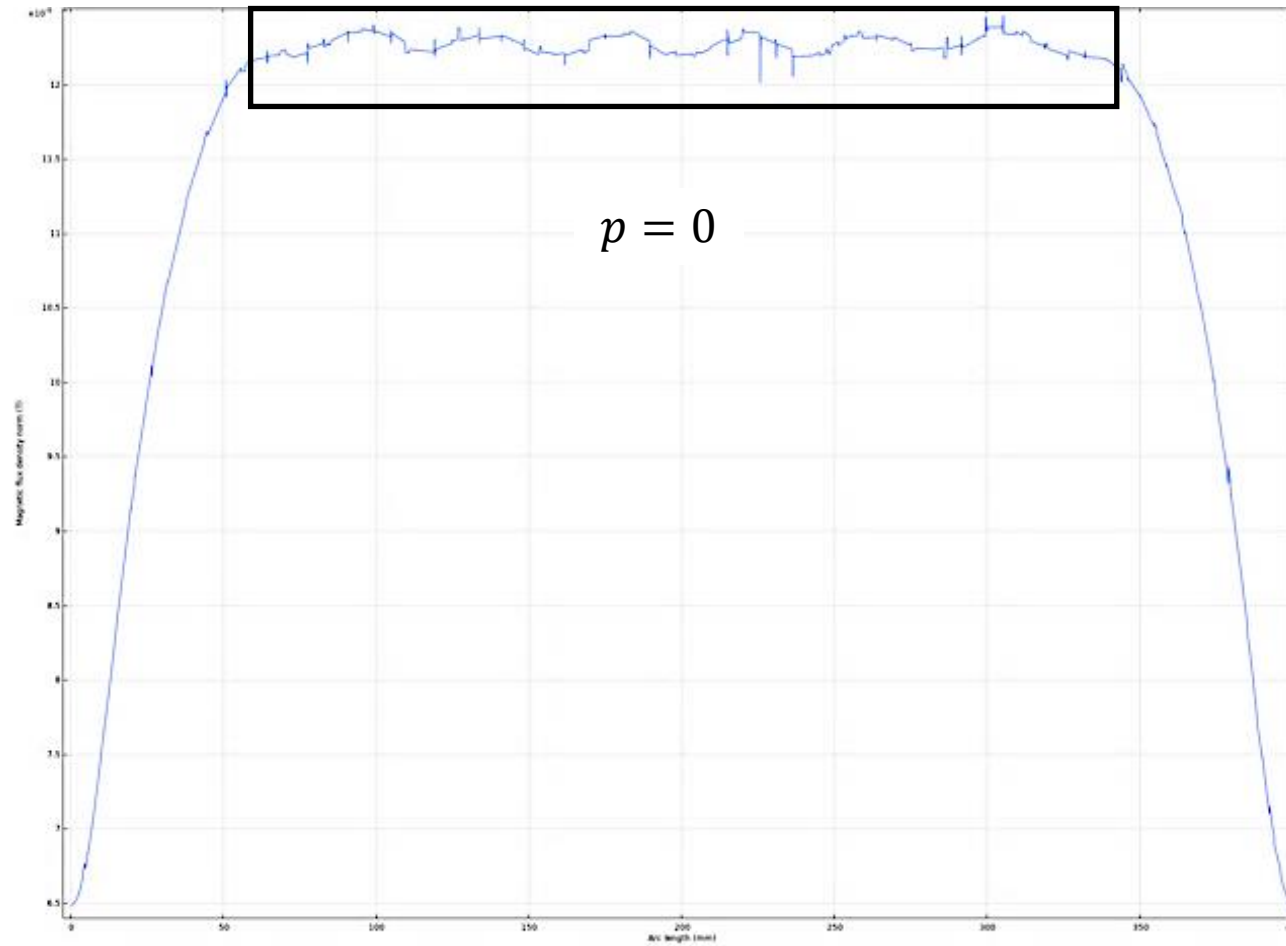
Center of Mass Positions

Magnet Set	Magnet Center of Mass Height	Gap Center of Mass Height
1	12.5	$33.\bar{3}$
2	$54.1\bar{6}$	75
3	$95.8\bar{3}$	$116.\bar{6}$
4	137.5	$158.\bar{3} + \frac{p}{2}$
5	$179.1\bar{6} + p$	200
6	$220.8\bar{3} - p$	$241.\bar{6} - \frac{p}{2}$
7	262.5	$283.\bar{3}$
8	$304.1\bar{6}$	325
9	$345.8\bar{3}$	$366.\bar{6}$
10	387.5	

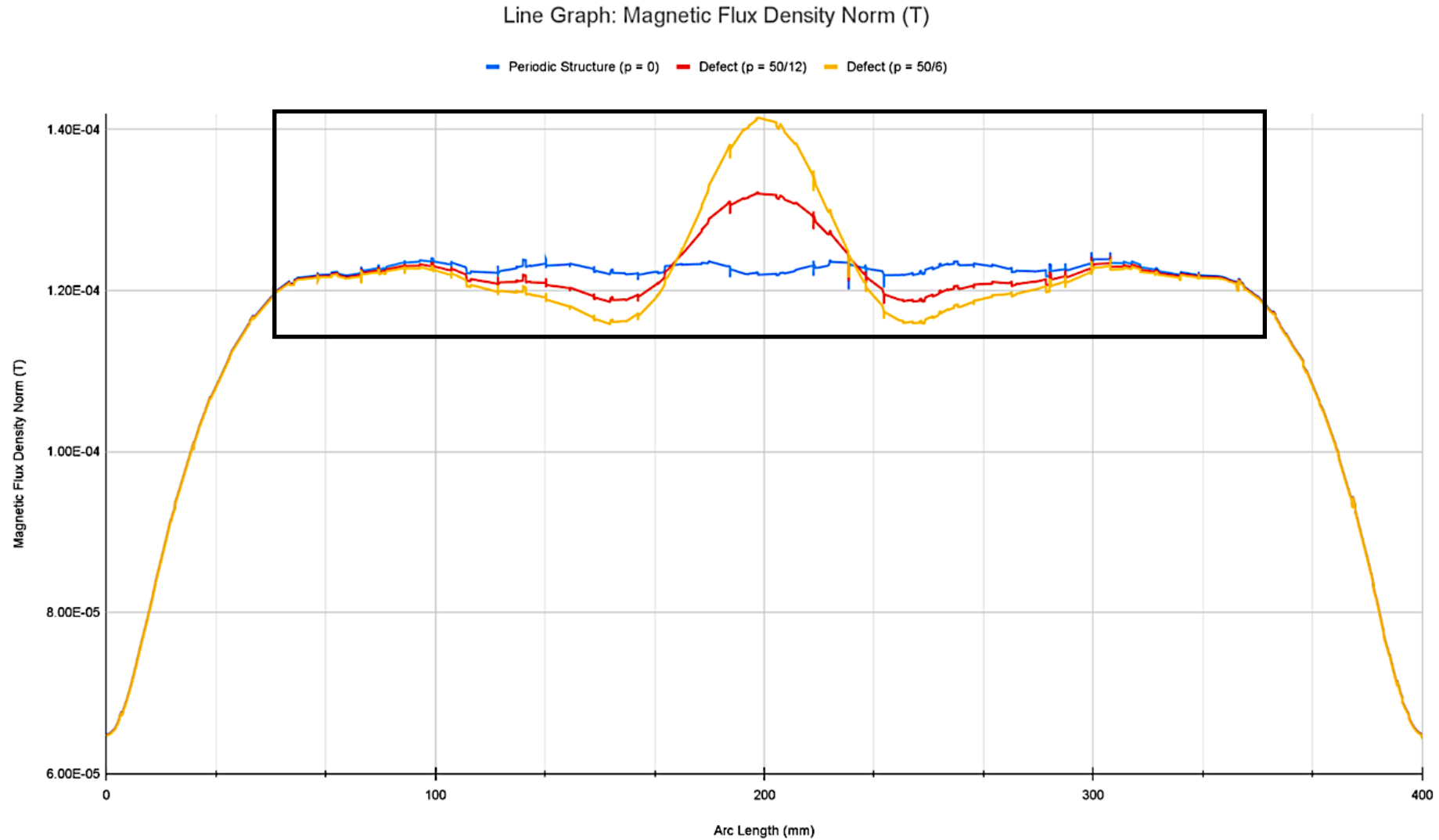
Topological Defect Positions



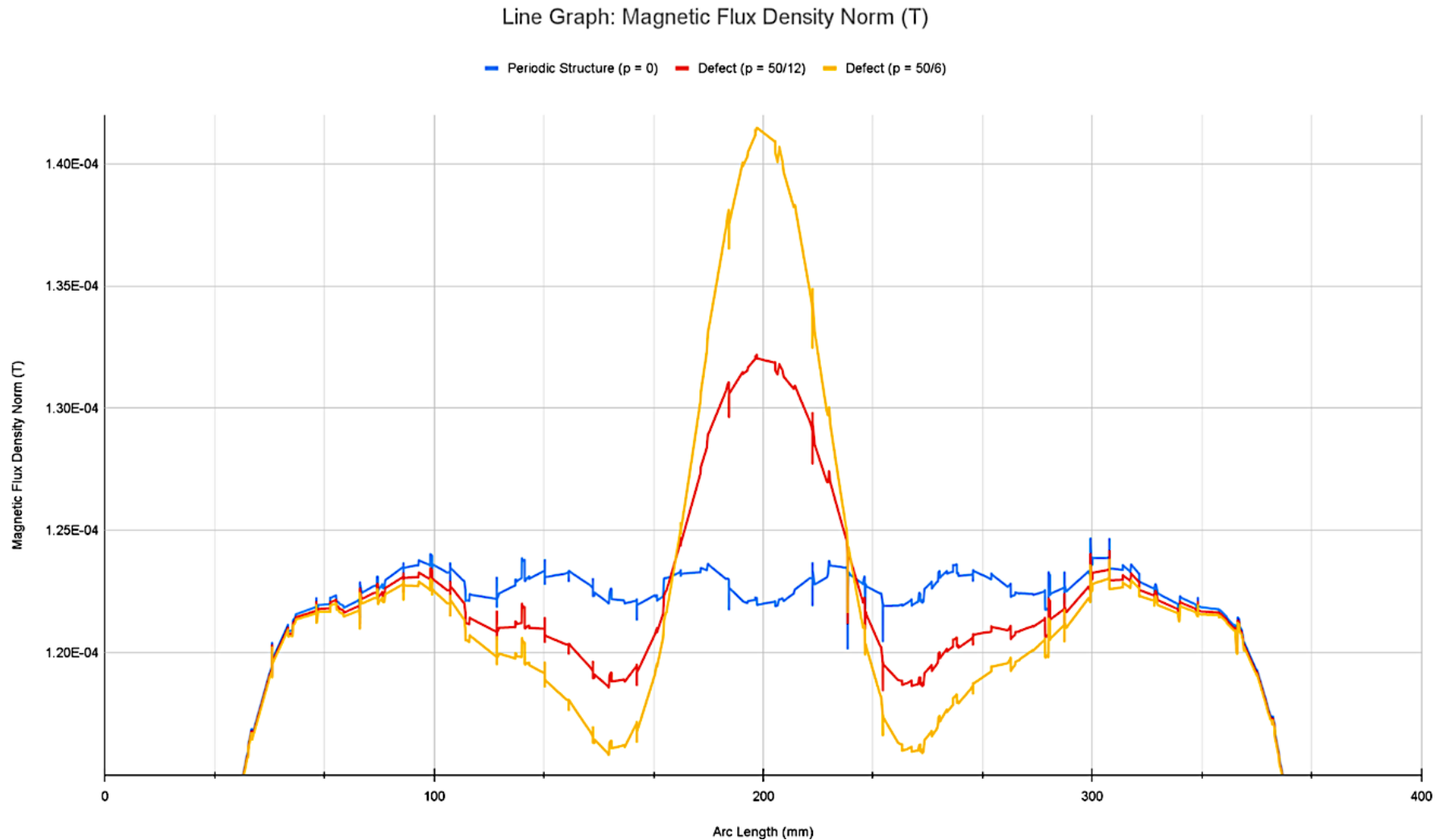
Periodic Structure Magnetic Flux Density



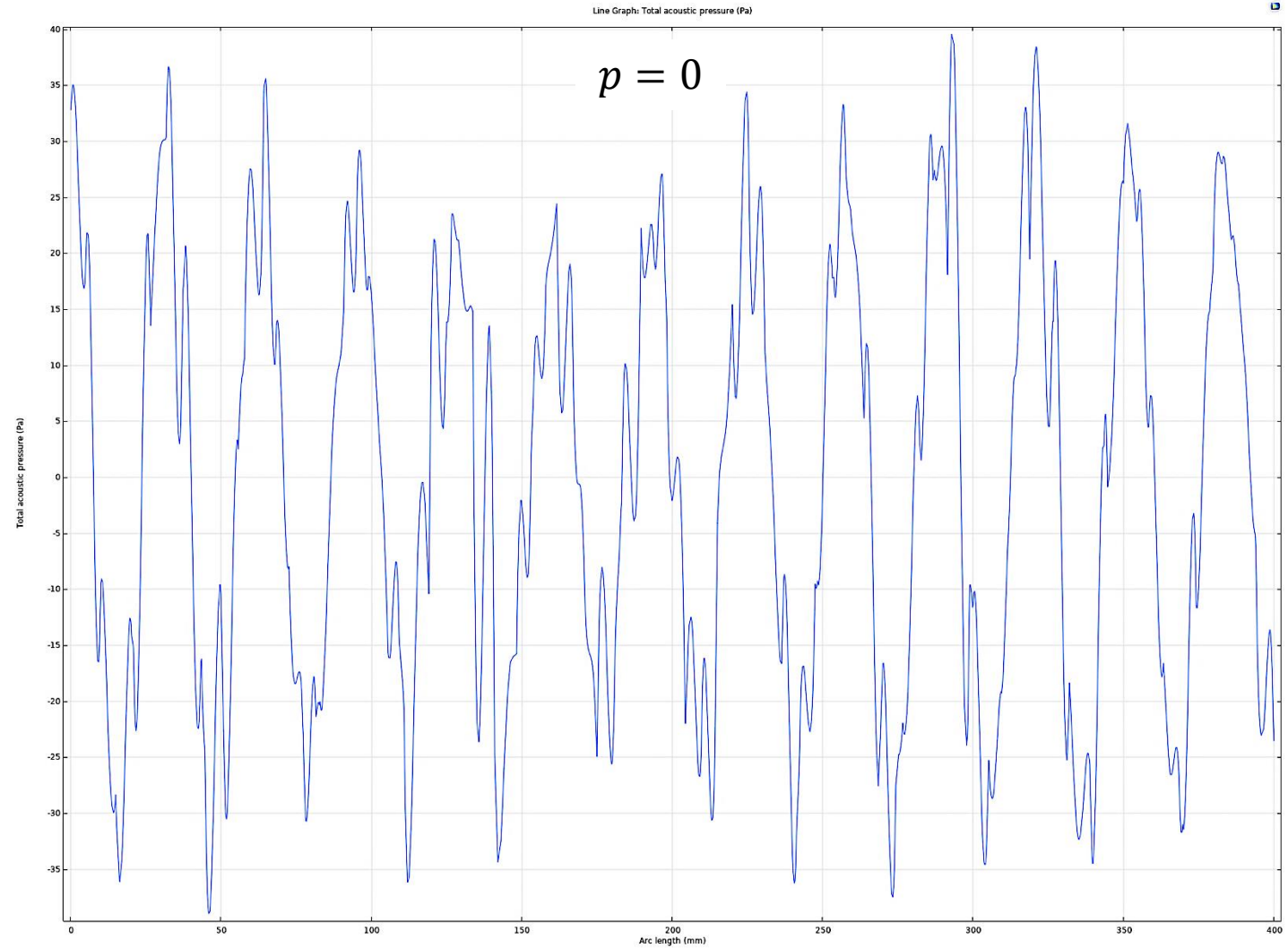
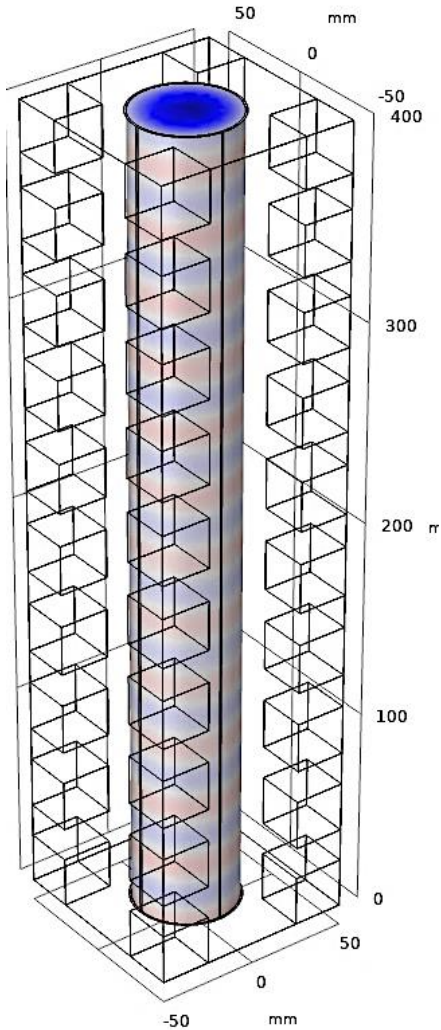
Magnetic Flux Densities



Magnetic Flux Densities



Periodic Structure Acoustic Pressure Plots



Acoustic Pressure Plots

